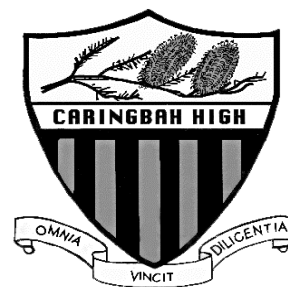


**Caringbah High School**  
**Year 11 Mathematics 2014**  
**Semester 2 Examination**



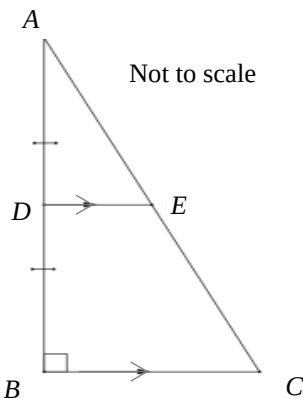
**Time Allowed: 2 hour + 5 minutes reading**

**Instructions**

- Start each question in a new booklet.
- Answers unsupported by working may not attract full marks.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- Attempts that show an understanding of concepts may attract some marks.

**Question 1 (13 marks)**

**Marks**

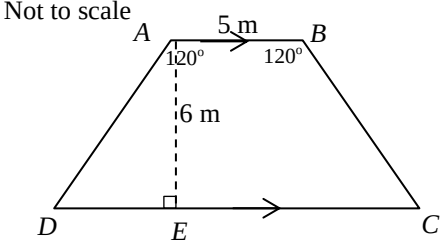
- a) Find the midpoint of the interval joining the points  $A(5, -2)$  and  $B(-3, -6)$ . 1
- b) State the gradient of the line  $3y - 2x + 5 = 0$ . 1
- c) Sketch the graph of  $y = 4 - x^2$ , showing all intercepts on axes. 2
- d)  The  $\triangle ABC$  is right-angled at  $B$ .  
 $BC \parallel DE$ .  
 $AD = DB$ .  
 i) Show that  $\triangle ADE$  is right-angled. 1  
 ii) Prove that  $\triangle ABC \sim \triangle ADE$ . 2
- e) Sketch the graph of  $y = 3^x$ . 1
- f) State the range of the function  $xy = 4$ . 1
- g) If  $f(x) = x^2 + 4x - 1$ , find all possible values for  $x$  if  $f(x) = 20$ . 2
- h) Find the derivative of  $3x^2 - 2x + 1$ . 2

## Question 2 (13 marks)

- a) The points  $A(-3, 2)$ ,  $B(5, 1)$ ,  $C(7, -2)$  and  $D$  form a parallelogram.
- i) Plot these points on a number plane and show the coordinates of  $D$ . 2
- ii) Find the coordinates of the point  $E$  where  $AC$  intersects with  $BD$ . 1
- b) Determine the natural domains of the following functions.
- i)  $y = \frac{x}{x+1}$  1
- ii)  $y = \sqrt{x-4}$  1
- c) Find the distance between the points  $P(5, 9)$  and  $Q(-3, 4)$  correct to 3 significant figures. 2
- d) Find the derivatives of the following functions
- i)  $y = 4x^{\frac{3}{5}}$  2
- ii)  $y = \frac{5}{2x^4}$  2
- iii)  $y = \sqrt[3]{4x-2}$  2

### Question 3 (13 marks)

Marks

- a)  *ABCD* is an isosceles trapezium with  $AB \parallel DC$  and  $\angle DAB = \angle CBA = 120^\circ$ . The perpendicular height  $AE$  is 6 m.  $AB = 5$  m.
- i) Show that  $DE = 2\sqrt{3}$  m 2
- ii) Find the area of the trapezium. 2

- b) Sketch the graph of these parabolas and find the coordinates of their vertex.

i)  $y = (x - 2)(x + 4)$  2

ii)  $y = 3x(6 - x)$  2

- c) Find the derivatives of these functions.

i)  $f(x) = (3x^2 + 5)^4$  2

ii)  $f(x) = (2x^3 - 7)^3(4x - 1)$ , in simplest factorised form. 3

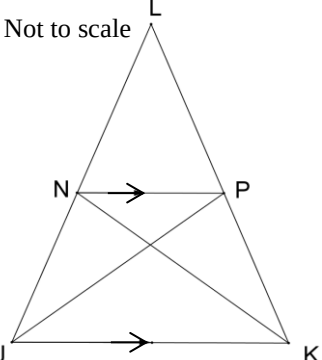
### Question 4 (13 marks)

- a) Find the equation of the line that passes through the points  $A(5, -2)$  and  $B(2, 1)$ . 3

- b) Find the point of intersection of the straight lines  $x + 2y = 5$  and  $x - y = 2$ . 2

c) i) Find  $\lim_{x \rightarrow -2} \frac{x^2 - 25}{x - 5}$  1

ii) Find  $\lim_{x \rightarrow 5} \frac{x^2 - 25}{x - 5}$  1

- d)  In the diagram,  $JKL$  is an isosceles triangle where  $JL = KL$  and  $JK$  is parallel to  $NP$ .

- i) Show that triangle  $LNP$  is an isosceles triangle. 2

- ii) Show that  $JN = KP$ . 1

- iii) Show that  $\triangle NJK \equiv \triangle PKJ$ . 3

**Question 5 (12 marks)****Marks**

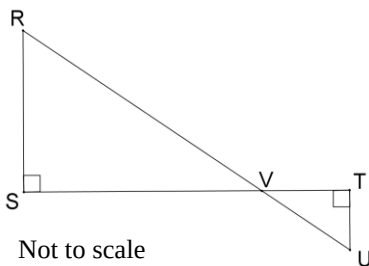
- a) State, with reasons, whether the function  $y = \sin x$  is an odd or an even function. 1
- b) Find the perpendicular distance from the point  $(2, 5)$  to the line  $3x + 4y = 1$ . 2
- c) Find the equation of the line through the point  $(-2, 6)$  that is parallel to the line  $2x - 4y + 3 = 0$ . 2
- d) Find the equation of the tangent to the curve  $y = x^3 - 2x^2 + 3x - 1$  at the point  $(1, 1)$ . 3
- e) Show that the line  $3x - 2y + 14 = 0$  is a tangent to the circle  $(x - 1)^2 + (y - 2)^2 = 13$ . 2
- f) Shade the region defined by  $y \leq x^3 + 1$  2

**Question 6 (12 marks)**

- a) Find the sum of the internal angles of a decagon. 1
- b) The exterior angle of a regular polygon is  $24^\circ$ . 2  
How many sides does the polygon have?
- c) i) On the same set of axes, sketch the graphs of  $y = |2x - 1|$  and  $y = 3$ . 2  
ii) Find the points of intersection of these two graphs. 2
- d) Differentiate from first principles  $y = x^2 - 3x + 1$ . 2
- e) Differentiate  $y = \frac{(3x - 2)^3}{5x + 1}$  3

**Question 7 (12 marks)**

**Marks**

- a)  2
- Rudi used similar triangles to measure the width of a creek. The field diagram shown shows two points, R and S, on opposite banks of the creek. Using line of sight, he constructed a smaller similar triangle VTU on the land.
- He measured the following distances.
- SV = 25 m, VT 5 m, TU = 12 m and VU = 13 m.
- Find the width of the creek RS.
- b) Sketch the graph of  $y = \sqrt{25 - x^2}$  2
- c) Find the equation of the line that passes through the origin and the point of intersection of the lines  $2x + 3y - 9 = 0$  and  $x - y - 2 = 0$ . 4
- d) A function is defined as  $f(x) = \begin{cases} x^3 & x < 0 \\ x + 1 & x \geq 0 \end{cases}$
- i) Explain why  $f(x)$  is not a continuous function. 1
- ii) Evaluate  $f(-2) + f(3)$ . 1
- iii) Sketch the graph of  $y = f(x)$ . 2

**Question 8 (12 marks)**

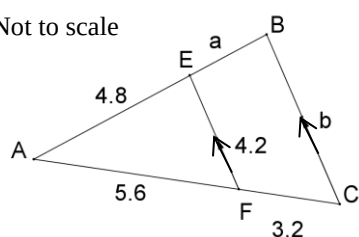
**Marks**

a) i) Differentiate  $y = \sqrt{x+3}$ . 2

ii) Find the gradient of the curve  $y = \sqrt{x+3}$  at the point (1, 2). 1

iii) Find the equation of the normal to the curve  $y = \sqrt{x+3}$  at the point (1, 2). 2

b) Not to scale In this diagram  $FE \parallel CB$ . 2



$AE = 4.8$ ,  $AF = 5.6$ ,  $FE = 4.2$  and  $FC = 3.2$ .

Find values for a and b.

c) Sketch the graph of  $y = \frac{-1}{x+2}$ , showing all asymptotes. 2

d) The normal to the curve  $y = ax^3 - 1$  at  $x = 1$  meets the positive  $x$ -axis at an angle of  $120^\circ$ . Find the value of  $a$ . 3

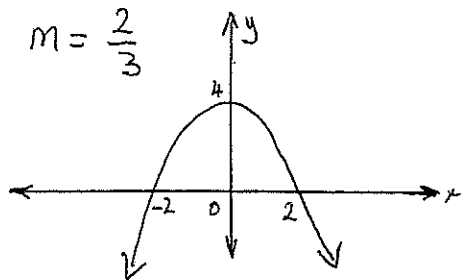
**END OF PAPER**

Question 1

a)  $(1, -4)$

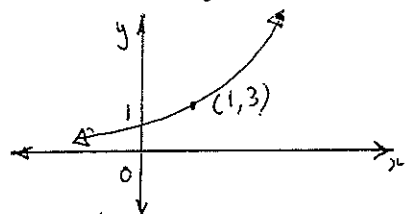
b)  $m = \frac{2}{3}$

c)



d)  $\angle ADE = \angle ABC$  corresponding angles,  $DE \parallel BC$   
 $\therefore 90^\circ$  - given  $\therefore \angle ADE$  is right-angled

e)



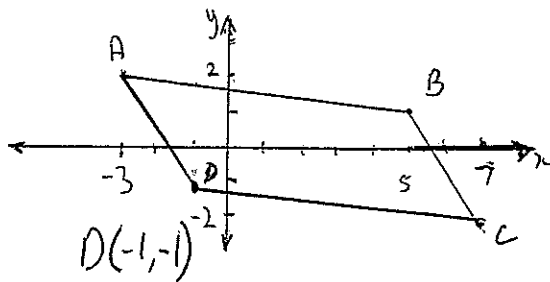
f) All real  $y$ ,  $y \neq 0$

g)  $x^2 + 4x - 1 = 20$   
 $x^2 + 4x - 21 = 0$   
 $(x+7)(x-3) = 0$   
 $x = -7$  or  $x = 3$

h)  $6x - 2$

Question 2

a) i)



ii)  $E(2, 0)$

b) i) D: All real  $x$ ,  $x \neq -1$

ii) D:  $x \geq 4$

c)  $d = \sqrt{(5+3)^2 + (9-4)^2}$   
 $= \sqrt{89}$   
 $= 9.43$

d) i)  $y' = \frac{12}{5} x^{-\frac{2}{5}}$

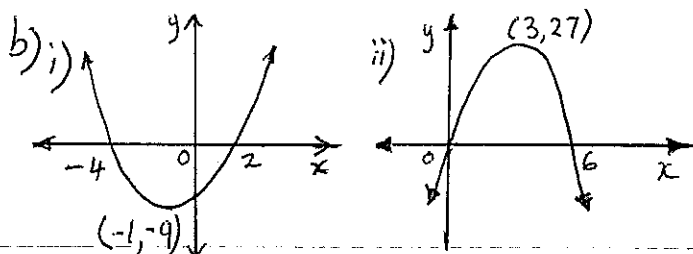
ii)  $y = \frac{5}{2} x^{-4}$   $y' = -10x^{-5}$   
 $= -\frac{10}{x^5}$

iii)  $y = (4x-2)^{\frac{1}{3}}$   
 $y' = \frac{1}{3} (4x-2)^{-\frac{2}{3}}$   
 $= \frac{1}{3 \sqrt[3]{(4x-2)^2}}$

Question 3

a) i)  $\angle DAE = 30^\circ$   $\tan 30^\circ = \frac{DE}{6}$   
 $\frac{1}{\sqrt{3}} = \frac{DE}{6}$   
 $DE = \frac{6}{\sqrt{3}}$   
 $= 2\sqrt{3}$

ii)  $A = \frac{h}{2}(a+b)$   
 $= \frac{6}{2}(5 + (5+4\sqrt{3})) m^2$   
 $= 30 + 12\sqrt{3} m^2$



c) i)  $f'(x) = 24x(3x^2+5)^3$

ii)  $u = (2x^3-7)^3$   $v = (4x-1)$   
 $u' = 18x^2(2x^3-7)^2$   $v' = 4$

$f'(x) = 18x^2(2x^3-7)^2(4x-1) + 4(2x^3-7)^3$   
 $= 2(2x^3-7)^2 [9x^2(4x-1) + 2(2x^3-7)]$   
 $= 2(2x^3-7)^2 (40x^3 - 9x^2 - 14)$

Question 4

a)  $m = \frac{3}{-3}$  Eqn  $(y-1) = -1(x-2)$   
 $= -1$   $y-1 = -x+2$   
 $x+y-3=0$

b)  $x+2y=5$  ①  
 $x-y=2$  ②  
 $① - ② \cdot 3y = 3$   
 $y = 1$   $x = 3$

Q4c

$$i) \frac{4-25}{-2-5} = \underline{\underline{3}}$$

$$ii) \lim_{x \rightarrow 5} \frac{(x-5)(x+5)}{(x-5)} = \underline{\underline{10}}$$

d)  $\angle LNP = \angle LJK$  corres.  $\angle$ s,  $NP \parallel JK$ .  
 $\angle LPN = \angle LKJ$  " " ,  $NP \parallel JK$ .  
 $\angle LJK = \angle LKJ$  angles opp. equal sides of an isos.  $\Delta$ .  
 $\therefore \angle LNP = \angle LPN$   
 $\therefore \Delta LNP$  is isosceles

ii)  $JL = KL$  given  
 $LN = LP$  equal sides of isos.  $\Delta$  ①

$$NJ = JL - LN$$

$$PK = KL - LP \quad JL - LN = KL - LP \quad \text{from ①}$$

$$\therefore NJ = PK.$$

iii) In  $\Delta NJK$  and  $\Delta PKJ$

S  $JK$  is common

A  $\angle NJK = \angle PKJ$  angles opp. equal sides of isos  $\Delta LJK$ .

S

$$NJ = PK \quad \text{from ii)}$$

$$\therefore \Delta NJK \equiv \Delta PKJ \quad \text{SAS}$$

### Question 5

a)  $\sin(-\theta) = -\sin \theta$  so  $\sin x$  is odd  
 NB. You must reference the question. ie use  $\sin x$ .

$$b) d = \left| \frac{(3 \times 2) + (4 \times 5) - 1}{\sqrt{3^2 + 4^2}} \right|$$

$$= \underline{\underline{5}}$$

$$c) m = \frac{1}{2} \quad (y-6) = \frac{1}{2}(x+2)$$

$$2y-12 = x+2$$

$$\underline{\underline{x-2y+14=0}}$$

d)  $y' = 3x^2 - 4x + 3$  Eqn of tangent  
 $x=1 \quad m=2$   
 $(y-1) = 2(x-1)$   
 $y-1 = 2x-2$   
 $\underline{\underline{2x-y-1=0}}$

e) Circle centre  $(1,2)$  rad =  $\sqrt{13}$

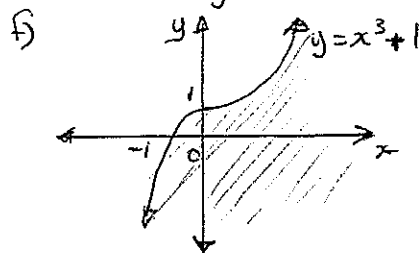
Using  $\perp$  dis from centre

$$d = \left| \frac{(3 \times 1) - (2 \times 2) + 14}{\sqrt{3^2 + (-2)^2}} \right|$$

$$= \left| \frac{13}{\sqrt{13}} \right|$$

$$= \sqrt{13} \quad (= \text{radius})$$

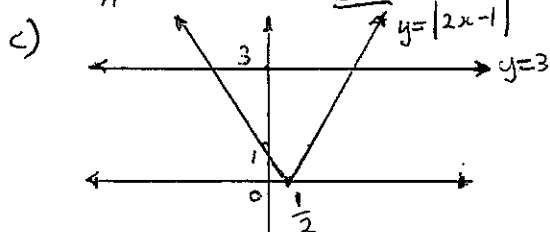
$\therefore 3x-2y+14=0$  is a tangent.



### Question 6

$$a) 180^\circ \times 8 = \underline{\underline{1440^\circ}}$$

$$b) \frac{360}{n} = 24 \quad \therefore n = \underline{\underline{15}}$$



$$ii) 2x-1=3 \quad \text{or} \quad 2x-1=-3$$

$$x = \underline{\underline{2}}$$

$$y = \underline{\underline{3}}$$

$$x = \underline{\underline{-1}}$$

$$y = \underline{\underline{3}}$$

$$d) f(x) = x^2 - 3x + 1$$

$$f(x+h) = (x+h)^2 - 3(x+h) + 1$$

$$= x^2 + 2xh + h^2 - 3x - 3h + 1$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2 - 3x - 3h + 1) - (x^2 - 3x + 1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + h - 3)}{h}$$

$$= \underline{\underline{2x-3}}$$

e)  $y' = \frac{vu' - uv'}{v^2}$   $u = (3x-2)^3$   $v = (5x+1)$   
 $u' = 9(3x-2)^2$   $v' = 5$

$$= \frac{9(3x-2)^2(5x+1) - 5(3x-2)^3}{(5x+1)^2}$$

$$= \frac{(3x-2)^2 [9(5x+1) - 5(3x-2)]}{(5x+1)^2}$$

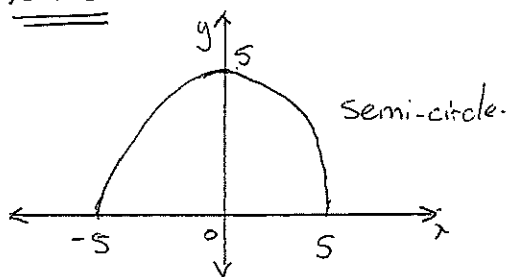
$$= \frac{(3x-2)^2 (5x+19)}{(5x+1)^2}$$

## Question 7

a)  $\frac{x}{25} = \frac{12}{5}$

$x = 60$

b)



c)

$L_1 + kL_2 = 0$

$(2x + 3y - 9) + k(x - y - 2) = 0$

$x=0, y=0 \Rightarrow -9 - 2k = 0$

$k = -4\frac{1}{2}$

$(2x + 3y - 9) - \frac{9}{2}(x - y - 2) = 0$

$x - 3y = 6$

(Alternatively, solving simultaneously to find point of intersection  $(3,1)$  giving  $m = \frac{1}{3}$ )

d) i) at  $x=0$   $\text{FOV} = 1$   
but as  $x \rightarrow 0$  from the negative.

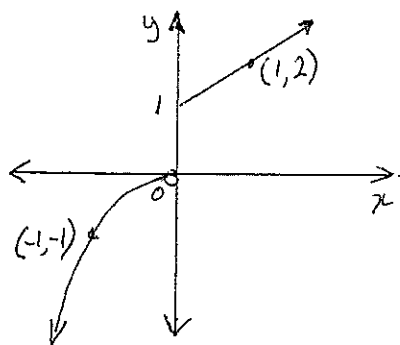
$\text{FOV} \rightarrow 0^3 = 0$

$\therefore$  There is a gap in the graph at  $x=0$

ii)  $f(-2) + f(3) = -8 + 4 = -4$

NB: You must reference the question on the paper.

iii)



## Question 8

a) i)  $y = (x+3)^{\frac{1}{2}}$

$y' = \frac{1}{2}(x+3)^{-\frac{1}{2}}$

$= \frac{1}{2\sqrt{x+3}}$

ii)  $x=1 \quad m = \frac{1}{4}$

iii)  $M_N = -4$  Eqn of normal

$(y-2) = -4(x-1)$

$y-2 = -4x+4$

$4x + y - 6 = 0$

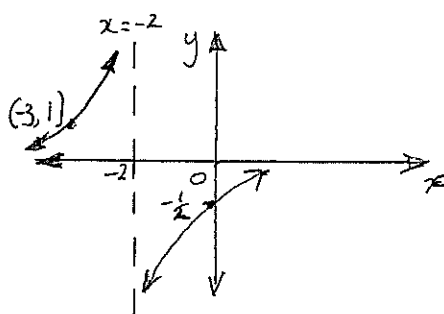
b)  $\frac{a}{4 \cdot 8} = \frac{3 \cdot 2}{5 \cdot 6}$  ratio of intercepts on parallel lines.

$a = \underline{2.74}$  (2d.p)

$\frac{b}{4 \cdot 2} = \frac{8 \cdot 8}{5 \cdot 6}$  ratio of sides of similar  $\Delta$ s.

$b = 6.6$

c)



d)  $y' = 3ax^2$

at  $x=1 \quad m_T = 3a \quad m_N = -\frac{1}{3a}$

$m_N = \tan 120^\circ$

$= -\sqrt{3}$

$-\frac{1}{3a} = -\sqrt{3}$

$a = \underline{\underline{\frac{1}{3\sqrt{3}}}} \quad \left(\frac{\sqrt{3}}{9}\right)$